

Shen Shu Yu Jin Wu: XR Virtual Production Empowering the Visual Spatial Construction and Immersive Experience of Chinese Classical Dance Films

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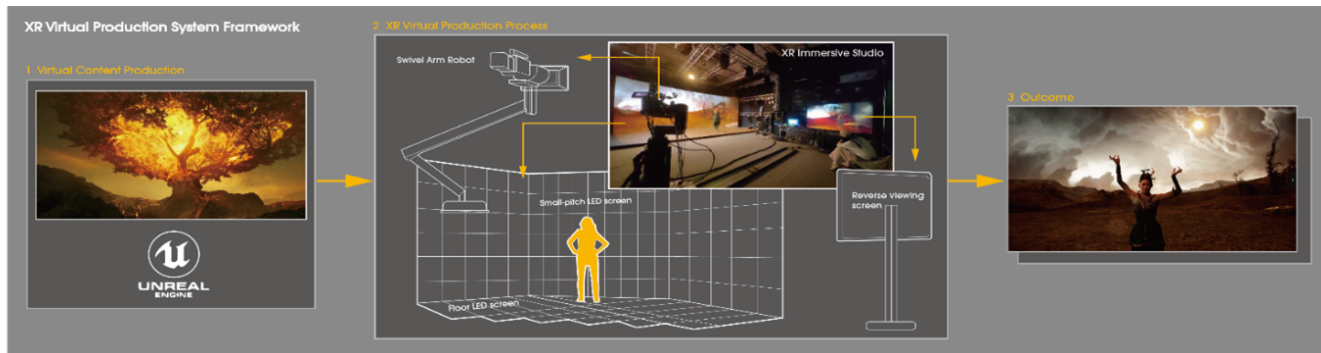


Figure 1: XR Virtual Production System Framework

Abstract

Extended Reality (XR) virtual production combines cutting-edge technologies such as real-time rendering, virtual reality, dynamic tracking, and artificial intelligence, fundamentally transforming traditional approaches to image creation and the aesthetics of artistic expression. By integrating XR virtual production techniques with Chinese classical dance imagery, this research introduces a new visual language and expressive form for traditional dance. The project not only explores innovative pathways for the evolution of Chinese classical dance but also opens up novel technological frameworks and dissemination methods.

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1 Introduction

In recent years, the growing popularity of concepts such as virtual production, the metaverse, and digital collectibles has propelled immersive audiovisual experiences—integrating real and virtual elements—into the cultural spotlight [Chen et al. 2024]. Chinese Classical Dance, rooted in traditional folk dance and refined by generations of practitioners, embodies distinct Chinese aesthetics. Under the lens of digital media, innovating and developing Chinese Classical Dance has become a complex and challenging process, requiring the strategic integration of new media capabilities. Dance-on-film has long served as a hybrid of choreography and cinematic language, enabling dance to transcend physical stages and function

as a cross-cultural medium for communication and storytelling [Yuan et al. 2025].

XR-based virtual production combines LED display systems, real-time rendering engines, high-precision camera tracking, and advanced compositing techniques to create dynamic virtual environments in real time. Actors perform against LED walls displaying fully rendered 3D scenes, seamlessly integrating physical performance with digital backdrops. This allows the final visual output to be captured directly on set, ensuring high-fidelity compositing without post-production delays. For audiences, XR reframes interaction across spatial, visual, and narrative dimensions, constructing a stable and immersive experiential framework. Figure 2 shows the diagram of the virtual production system.

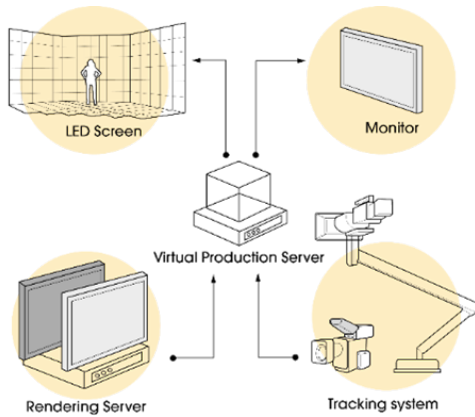


Figure 2: Virtual production system framework diagram

2 Design Approach

The XR-based film *Shen Shu Yu Jin Wu* draws inspiration from the ancient Chinese myth of the three-legged golden crow in *The Classic of Mountains and Seas: Eastern Sea Chapter*. This work integrates the graceful, intricate movements of Chinese classical dance with the spatial-temporal fluidity of virtual worlds, presenting a mythological narrative within an immersive stage that blurs reality and illusion. Unlike traditional green-screen setups, performers dance within an LED volume that emits its own light, creating a vivid and interactive environment. These LED screens replicate the designed landscapes and nuanced colour layers in real-time, allowing the dancers to fully inhabit their roles and enhancing the sense of presence. The resulting experience transports audiences on a mysterious journey to a mythical past, where physical movement and digital scenography converge to form a narrative space rich in dramatic and visual tension as shown in Figure 3.

2.1 Conceptual Design and Spatial Construction

In the final film, the story unfolds as follows: a mythical tree, the *Fu Sang Sacred Tree*, is awakened by a cacophony of sounds. The tree is home to nine divine birds that have accompanied it throughout the ages. Under the illumination of this radiance, light and heat burn the earth like fierce flames, ultimately giving birth to a phoenix through its rebirth. This narrative is broken down into three distinct scenes:

the sleeping *Fusang Sacred Tree*, the dialogue between the tree and the divine birds, and the phoenix's rebirth under the radiant light. The design of the main scene elements was accomplished using 3D modelling, with terrain editing and lighting design carried out in the real-time rendering engine *Unreal Engine*, preparing for the subsequent XR virtual shooting as shown in Figure 4.



Figure 3: Final XR production effect (Real time generation of real dancers and virtual scenes)



Figure 4: Virtual environment design and spatial design

2.2 Stage Performance and Virtual Shooting

XR technology enables the real-time extension of narrative space through the expansion of virtual environments. It merges the live performance of actors with augmented reality (AR) foreground effects, which is its biggest distinction from traditional film techniques. By using camera tracking and real-time image rendering technology, XR seamlessly integrates the constructed virtual scenes with the real-time scenes captured by the camera. This virtual shooting method overcomes the limitations of traditional green screen technology, providing greater creative freedom for content creation and performance. It immerses the actors in a visual, sensory, and interactive virtual digital environment, enhancing their artistic expression and emotional impact.

3 CONCLUSION

Through in-depth interviews with over 50 users, including choreographers, dancers, and audiences, positive feedback was collected across multiple dimensions such as performance content, performance experience, the relationship between audience and performers, and communication effects. They believe this creative approach offers an innovative path for immersive storytelling in dance. This technology continues to demonstrate its efficiency, convenience, and strong practical capabilities across various fields. It further extends the influence and creativity of digital technology, allowing more cultural stories and intangible heritage experiences to enter daily life in more imaginative ways, offering diverse audiovisual experiences. This brings an innovative digital development path for the inheritance and protection of traditional cultures.

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